

AI60203: FOUNDATIONS OF MACHINE LEARNING



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Course Components: LTP: 3-0-0

Classes

- Time table: Monday – 11:00-11:55 AM, Tuesday – 8:00-9:55 AM
- Venue: CRR 501 Class room
- Office Hours: Thursday, 5-6 PM, DoAI: [Reserve](#) (3 times)

Assignments

- Tutorial Questions – Take-home
- Mini-Project – More on this later

Evaluation

- Class Test - 1
- Mid-Sem Exam
- Class Test - 2
- End-Sem Exam

Grading Policy

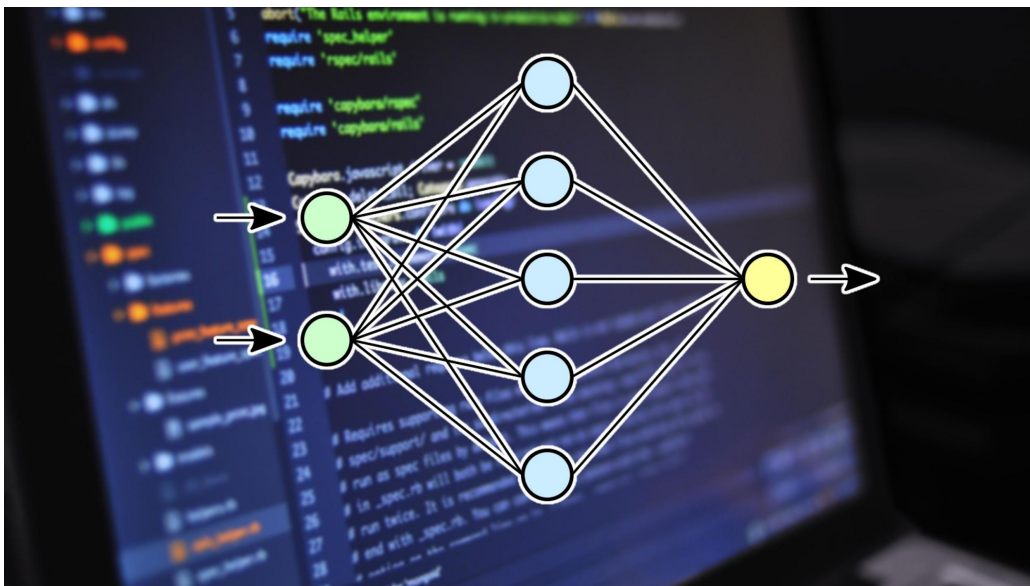
Theory - 75 %

- Evaluation: Class Test-1, Midsem, Class Test-2, and Endsem
 $7.5\% + 30\% + 7.5\% + 30\%$

Mini-project + In-class participation - 25 %

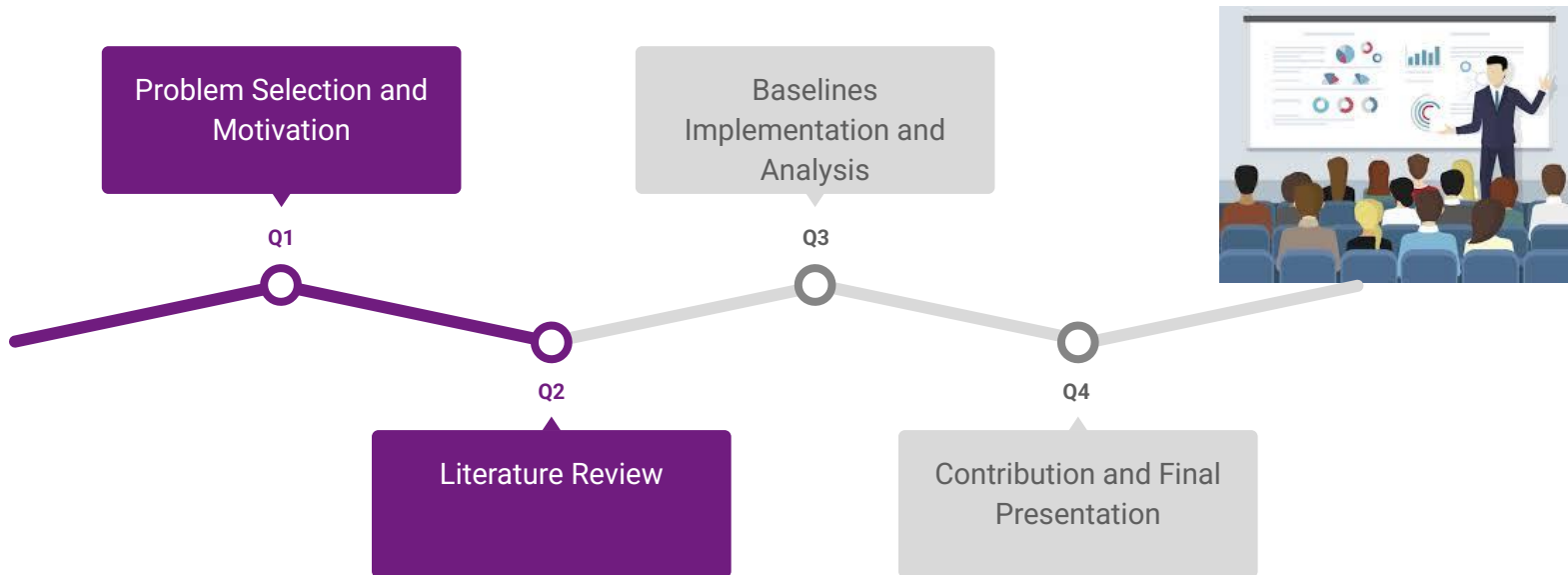
- Project evaluation in 2-3 Stages (distributed over the course)
- Answering questions, active participation, office hours, etc

Role of Creative-thinking + Implementation



1. Learning to think and code will empower you (especially in the AI era)
2. Job market or Academia still need Human Developers
3. There's no better time than to learn parallel with Theoretical knowledge.

Mini-Project



Philosophy



Human Creativity over AI

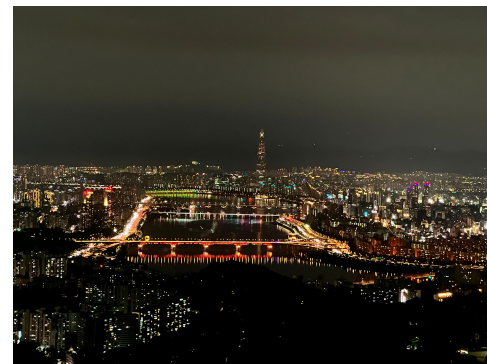


Me explaining my new groundbreaking idea



My advisor waiting to tell me it was already published

Best Case Scenario



Course Repository



Institute Moodle

1. **Availability of the lecture materials and References**
2. Tutorial Questions
3. Assignment submission
4. Announcements

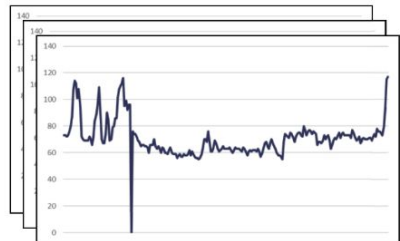
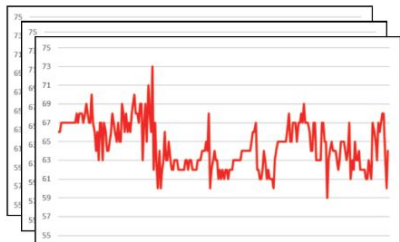
Important Links

Moodle IIT KGP - [Link](#)

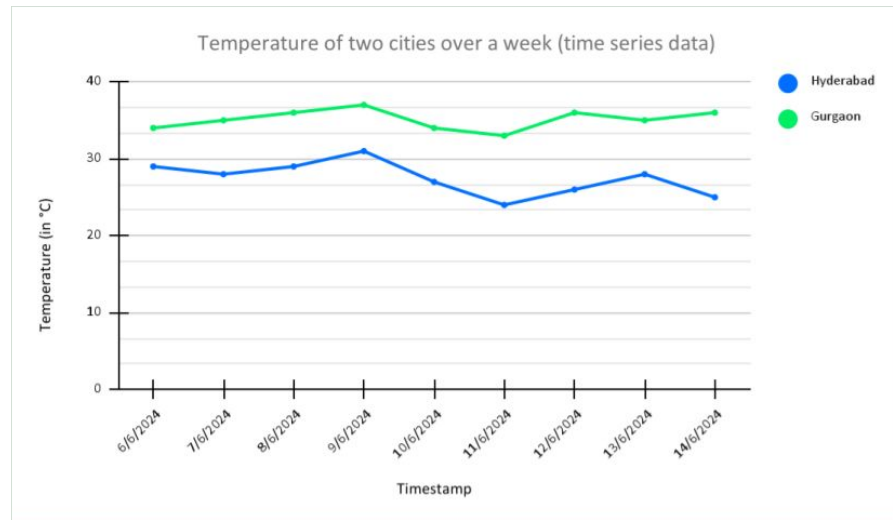
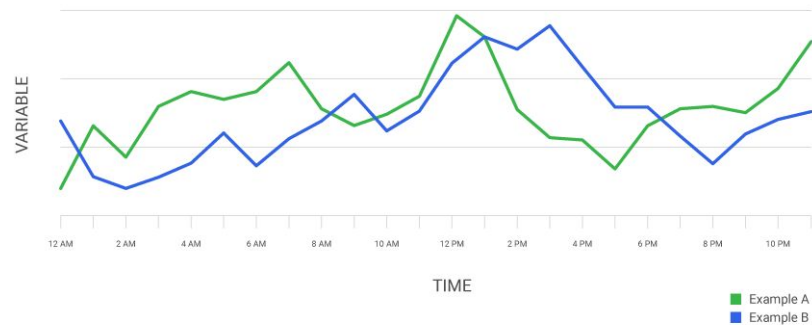
Reserve Office Hour Slot for Thursday - [Reserve](#)

Course Project Team - Project based

Time Series Data (1 dimensional)



IoT device



Time Series Forecasting

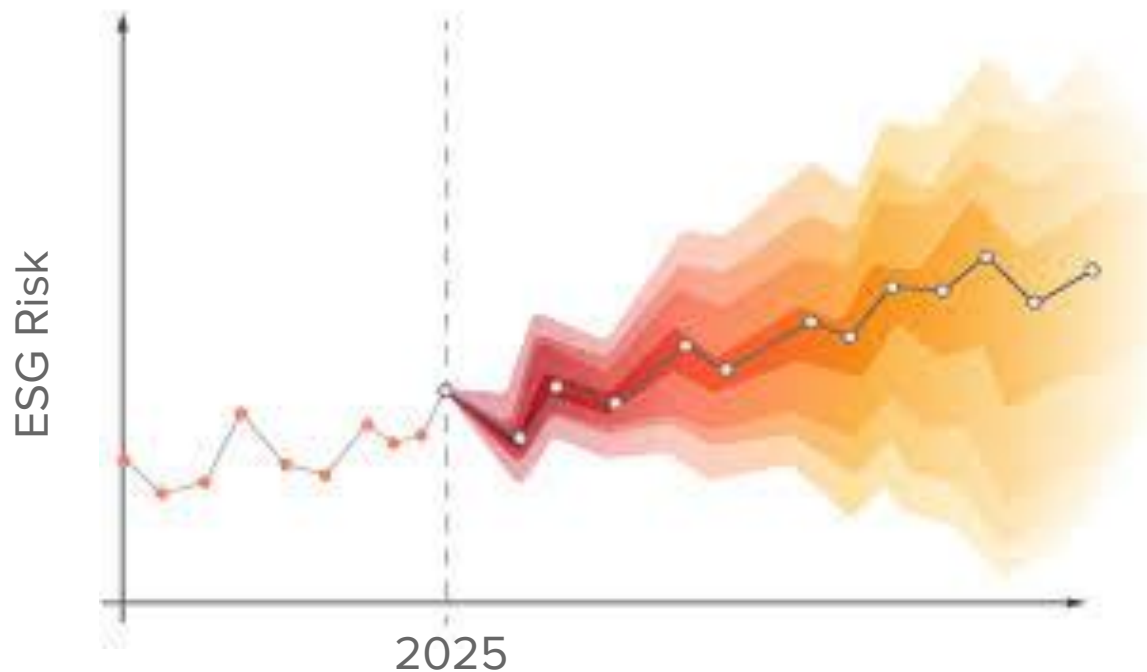
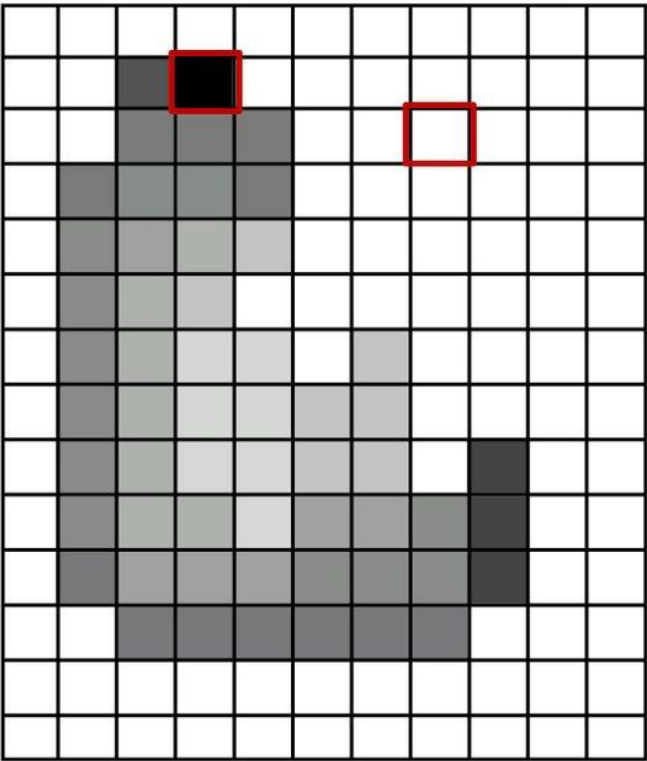
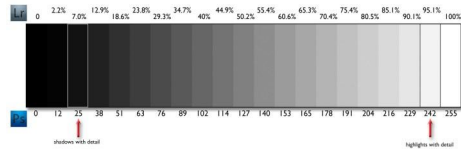


Image Data (2 dimensional)

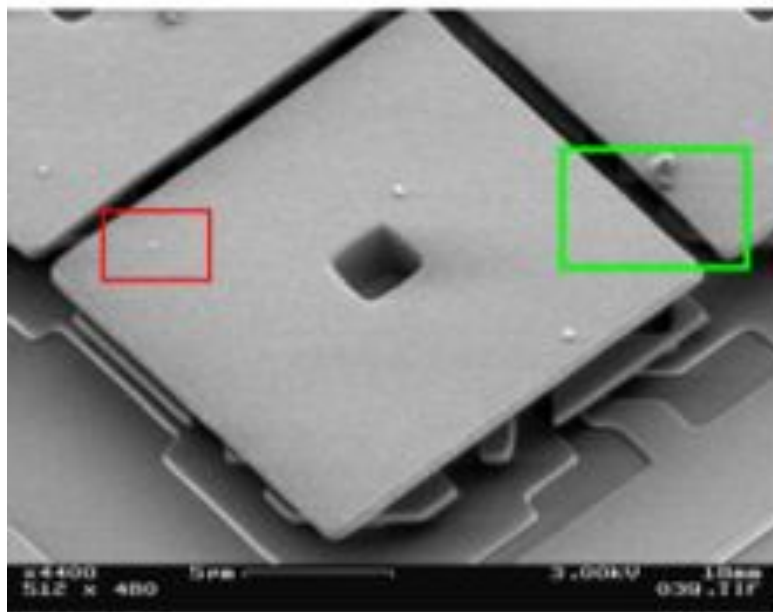


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255	255	255	255	255	255	255	255	255	255	255
255	255	20	0	255	255	255	255	255	255	255
255	255	75	75	255	255	255	255	255	255	255
255	75	95	95	75	255	255	255	255	255	255
255	96	127	145	175	255	255	255	255	255	255
255	127	145	175	175	175	255	255	255	255	255
255	127	145	200	200	175	175	95	255	255	255
255	127	145	200	200	175	175	95	47	255	255
255	127	145	145	175	127	127	95	47	255	255
255	74	127	127	127	95	95	95	47	255	255
255	255	74	74	74	74	74	74	255	255	255
255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255	255	255	255

0 = black; 255 = white

Gray Scale Image Analysis



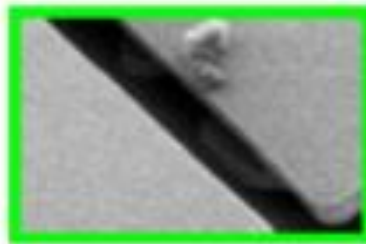
(a) Input



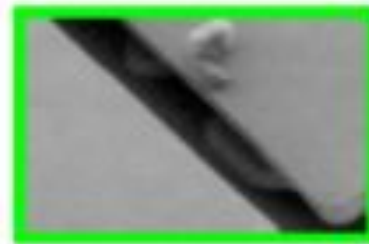
(b)



(c)



(d)

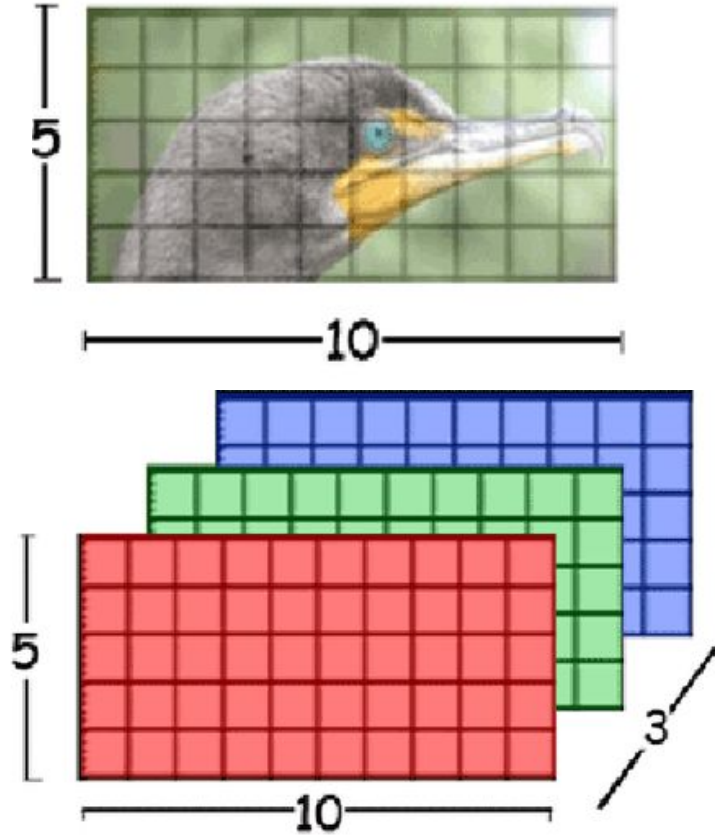
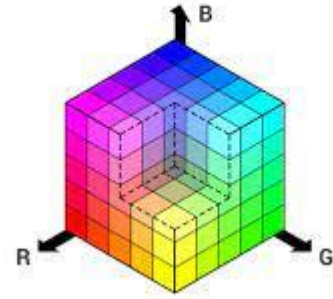


(e)

Input patches

Output patches

Color Image (3 dimensional)



Matlab RGB Matrix

Color name	RGB triplet	Color
Red	(255, 0, 0)	
Lime	(0, 255, 0)	
Blue	(0, 0, 255)	
White	(255, 255, 255)	
Black	(0, 0, 0)	
Gray	(128, 128, 128)	
Fuchsia	(255, 0, 255)	
Yellow	(255, 255, 0)	
Aqua	(0, 255, 255)	
Silver	(192, 192, 192)	
Maroon	(128, 0, 0)	
Olive	(128, 128, 0)	
Green	(0, 128, 0)	
Teal	(0, 128, 128)	
Navy	(0, 0, 128)	
Purple	(128, 0, 128)	

Color Image Analysis



(a) I/p (b) Blurred: O/p (c) Deblurred: O/p

(1) Semantic Segmentation



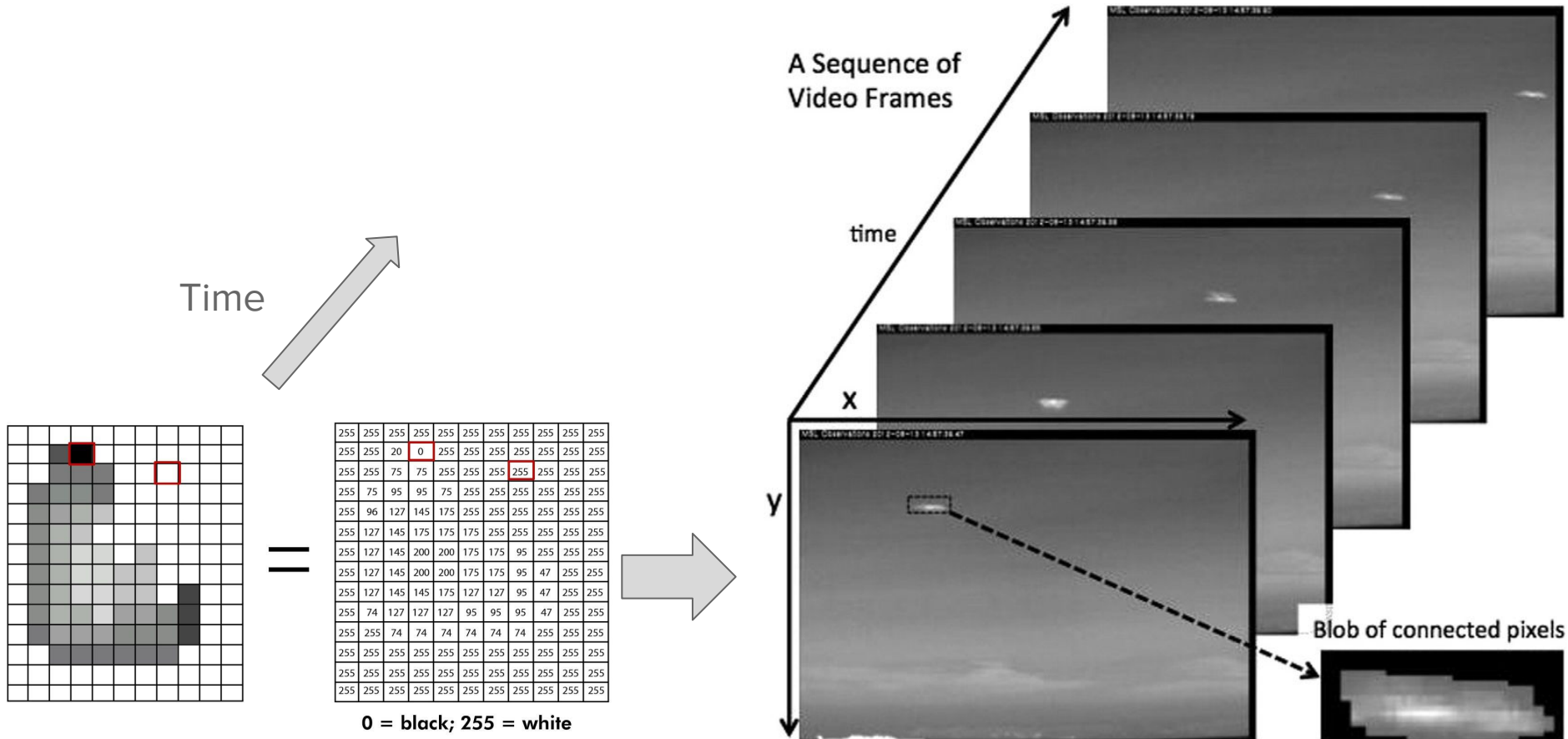
(a) Blurred: O/p



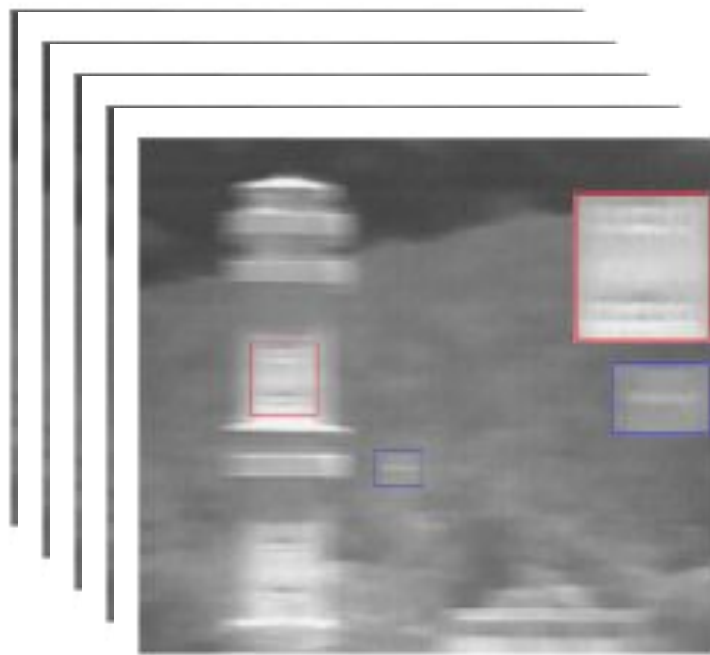
(a) Deblurred: O/p

(2) Object Detection

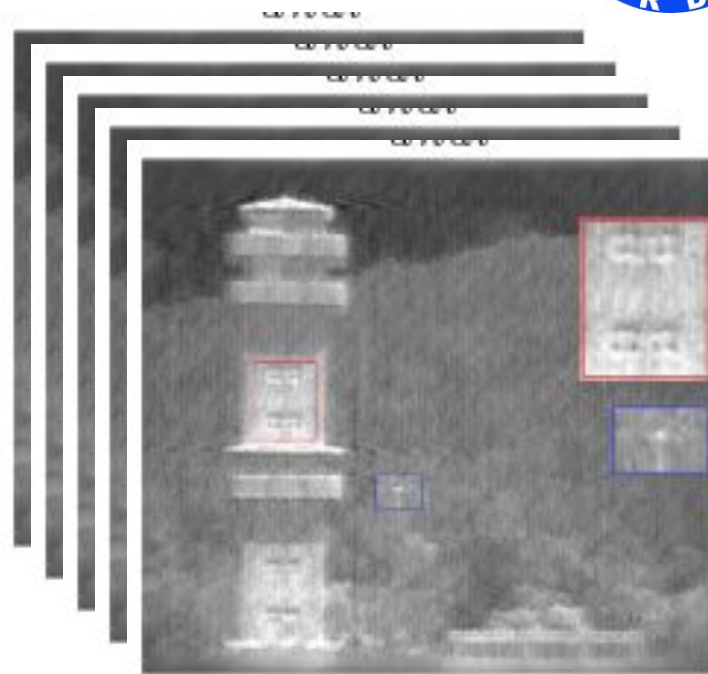
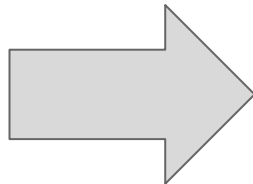
Grayscale Video Data (3 dimensional)



Grayscale Video Analysis

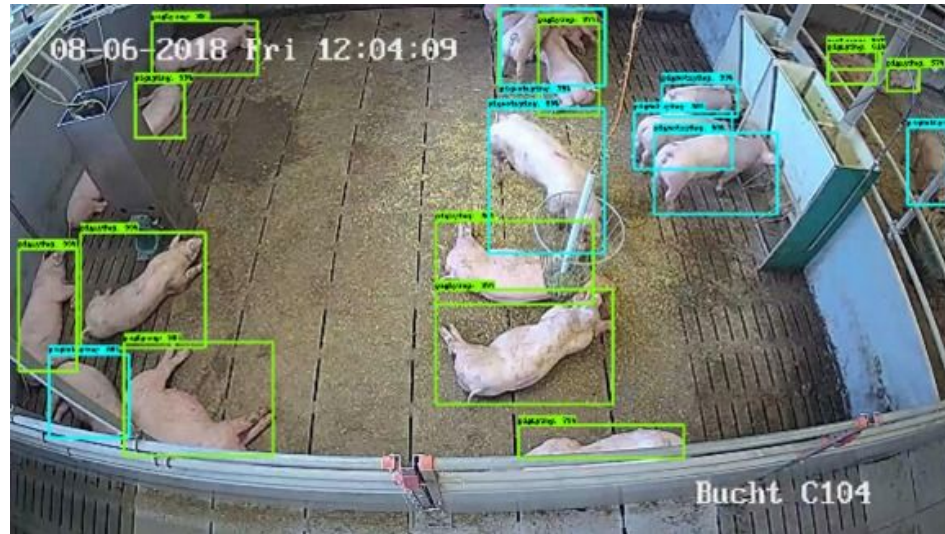


Blurred



3(c) $k_{anal}+WF$

Color Video Analysis



(a) Frame 8230



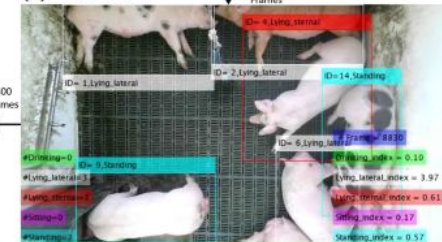
(b) Frame 8530



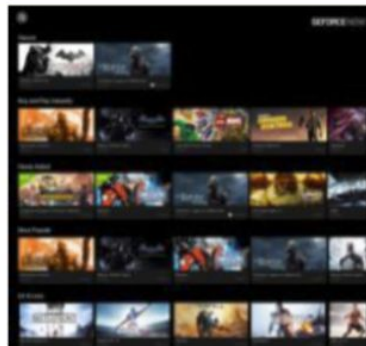
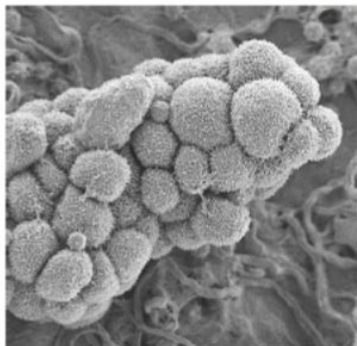
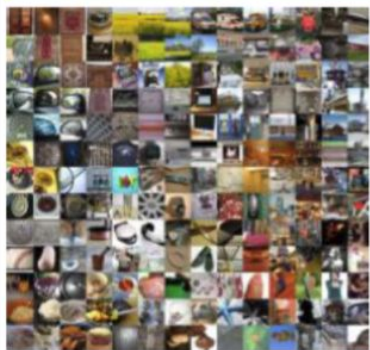
(d) Frame 9130



(c) Frame 8830



DEEP LEARNING EVERYWHERE



INTERNET & CLOUD

Image Classification
Speech Recognition
Language Translation
Language Processing
Sentiment Analysis
Recommendation

MEDICINE & BIOLOGY

Cancer Cell Detection
Diabetic Grading
Drug Discovery

MEDIA & ENTERTAINMENT

Video Captioning
Video Search
Real Time Translation

SECURITY & DEFENSE

Face Detection
Video Surveillance
Satellite Imagery

AUTONOMOUS MACHINES

Pedestrian Detection
Lane Tracking
Recognize Traffic Sign